

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Capital Cost Estimate (Class 4)

Document No.	CFU-000-PM-EST-001
Revision	A
Status	Issued for review (FEED)
Date	2026-10-02
Prepared	Claude Code
Checked / Approved	- / -

1 Basis

- AACE Class 4 (expected accuracy -30 % / +50 %), ISBL only, US Gulf Coast, 2026 USD (CEPCI 820, assumed).
- Purchased equipment: Towler & Sinnott correlations (CEPCI 532.9), with material factors per equipment metallurgy (5Cr 1.4, 9Cr 1.7, clad 1.6, Monel 1.9). Vessel shell weights come from the mechanical calculations (data/mech.json) where available (not available: estimated from design pressure and dimensions).
- Installation: Towler fluids-processing factors (erection 0.3, piping 0.8, instrumentation 0.3, electrical 0.2, civil 0.3, structures 0.2, lagging/paint 0.1).
- Excluded: Tankage, SWS, flare stack, CCR, main substation, OSBL racks; owner's costs, licence fees, land, escalation beyond 2026, catalyst/chemicals first fill.

2 Summary

Item	USD million
Purchased equipment (2026 USGC)	57.7
Installed ISBL (Towler factors)	170.4
Design & engineering (25 %)	42.6
Contingency (20 %)	42.6
Total installed cost, ISBL	255.7

Specific cost: **2,557 USD per BPSD** of capacity, ISBL. Check this against the owner's own benchmark data for grassroots CDU/VDU units before using it for budgeting. The factored method tends to under-estimate large, alloy-heavy refinery units.

3 Purchased equipment by type

Type	USD million	Share
Fired heater	19.24	33%
Column	11.93	21%
Air cooler	9.00	16%
Shell & tube	7.48	13%
Desalter	4.06	7%
Pump	3.60	6%
Package	0.74	1%
Drum	0.61	1%
Ejector	0.42	1%
Fan	0.33	1%
Air preheater	0.24	0%

4 Equipment detail

Tag	Service	Basis	Ce 2026 kUSD
C-101	Atmospheric crude fractionator	shell 188 t, 41 trays	4,212
C-102	Kero side stripper	shell 5 t, 6 trays	110
C-103	Diesel side stripper	shell 6 t, 6 trays	188
C-104	Ago side stripper	shell 4 t, 6 trays	141
C-105	Naphtha stabiliser (debutaniser)	shell 17 t, 19 trays	285
C-106	Naphtha splitter	shell 40 t, 38 trays	681

Tag	Service	Basis	Ce 2026 kUSD
C-201	Vacuum column (wet, packed)	shell 155 t, 20 trays, 308 m3 packing	6,310
H-101	Atmospheric crude charge heater	60.7 MW box	14,455
H-201	Vacuum heater	14.7 MW box	4,788
K-101A/B	H-101 forced-draft fan	2 x 90 kW	125
K-102A/B	H-101 induced-draft fan	2 x 200 kW	208
E-120	H-101 cast-iron/glass-tube air preheater	500 m2, 1 shell(s)	236
E-101	Crude / Top pumparound	359 m2, 1 shell(s)	174
E-102	Crude / Kerosene product	438 m2, 1 shell(s)	208
E-103	Crude / LVGO pumparound/product	804 m2, 2 shell(s)	386
E-104	Crude / Diesel product	378 m2, 1 shell(s)	183
E-105	Crude / Vacuum residue	431 m2, 1 shell(s)	288
E-106	Crude / Middle (kero) pumparound	1001 m2, 2 shell(s)	661
E-107	Crude / Diesel product	324 m2, 1 shell(s)	224
E-108	Crude / HVGO pumparound/product	2196 m2, 4 shell(s)	1,755
E-109	Crude / AGO product	263 m2, 1 shell(s)	231
E-110	Crude / Bottom (diesel) pumparound	886 m2, 2 shell(s)	716
E-111	Crude / Vacuum residue	1094 m2, 2 shell(s)	875
E-113	BPA / MP steam generator (kettle)	223 m2, 1 shell(s)	120
E-201	VR / LP steam generator (kettle)	648 m2, 1 shell(s)	425
E-114	Stabiliser feed / bottoms	117 m2, 1 shell(s)	82
E-115	Atm. overhead trim condenser	740 m2, 2 shell(s)	358
E-116	Stabiliser reboiler (HP steam)	190 m2, 1 shell(s)	108
E-117	Splitter reboiler (MP steam)	225 m2, 1 shell(s)	121
E-118	Desalter wash water / brine	84 m2, 1 shell(s)	71
E-202	1st-stage ejector intercondenser	549 m2, 1 shell(s)	335
E-203	2nd-stage ejector intercondenser	79 m2, 1 shell(s)	90
E-204	Ejector aftercondenser	30 m2, 1 shell(s)	72
A-101	Atm. overhead condenser	1667 m2 bare	2,976
A-103	Kero cooler	184 m2 bare	388
A-104	Diesel cooler	430 m2 bare	804
A-105	Ago cooler	287 m2 bare	562
A-106	Stabiliser overhead condenser	214 m2 bare	439
A-107	Splitter overhead condenser	881 m2 bare	1,568
A-108	Heavy naphtha product cooler	401 m2 bare	756
A-201	Lvgo cooler	397 m2 bare	748
A-202	Hvgo Product cooler	403 m2 bare	758
D-101A	Electrostatic desalter, 1st stage	shell 108 t	2,029
D-101B	Electrostatic desalter, 2nd stage	shell 108 t	2,029
D-102	Atm. overhead reflux drum (3-phase)	shell 19 t	219
D-103	Fuel gas knock-out drum	shell 1 t	31
D-105	Stabiliser reflux drum	shell 4 t	71

Tag	Service	Basis	Ce 2026 kUSD
D-106	Splitter reflux drum	shell 9 t	130
D-201	Ejector hotwell / sour water separator	shell 2 t	51
D-202	Vacuum off-gas knock-out drum	shell 0 t	25
D-104	Flare knock-out drum (unit)	shell 5 t	84
J-201	Steam ejector (20 -> 130 mbar(a))	2 x 50 % train	138
J-202	Steam ejector (120 -> 350 mbar(a))	2 x 50 % train	138
J-203	Steam ejector (330 -> 1100 mbar(a))	2 x 50 % train	138
P-101A/B	Crude charge	2 x 730 m3/h, 710 kW	457
P-102A/B	Desalted crude booster	2 x 805 m3/h, 710 kW	470
P-103A/B	Atm. reflux	2 x 159 m3/h, 55 kW	114
P-104A/B	Unstabilised naphtha	2 x 180 m3/h, 160 kW	171
P-105A/B	Overhead sour water	2 x 14 m3/h, 5.5 kW	45
P-106A/B	TPA pumparound	2 x 293 m3/h, 110 kW	164
P-107A/B	MPA pumparound	2 x 387 m3/h, 160 kW	231
P-108A/B	BPA pumparound	2 x 318 m3/h, 132 kW	205
P-109A/B	Kero product	2 x 122 m3/h, 55 kW	125
P-110A/B	Diesel product	2 x 151 m3/h, 75 kW	143
P-111A/B	Ago product	2 x 84 m3/h, 45 kW	110
P-112A/B	Atm. residue / vacuum heater charge	2 x 398 m3/h, 250 kW	268
P-114A/B	Desalter wash water	2 x 37 m3/h, 18.5 kW	65
P-115A/B	Stabiliser reflux / LPG	2 x 39 m3/h, 18.5 kW	66
P-116A/B	Splitter reflux / LN	2 x 146 m3/h, 55 kW	112
P-117A/B	Heavy naphtha product	2 x 142 m3/h, 45 kW	104
P-118	Desalter mud-wash / recycle	1 x 19 m3/h, 7.5 kW	24
P-201A/B	LVGO pumparound / product	2 x 198 m3/h, 110 kW	171
P-202A/B	HVGO pumparound / product	2 x 391 m3/h, 200 kW	248
P-203A/B	Slop wax	2 x 12 m3/h, 5.5 kW	55
P-204A/B	Vacuum residue (incl. quench)	2 x 196 m3/h, 132 kW	182
P-205A/B	Hotwell sour water	2 x 13 m3/h, 3.7 kW	41
P-206A/B	Hotwell slop oil	2 x 1 m3/h, 0.75 kW	31
X-101	Demulsifier injection package	vendor package	185
X-102	Caustic injection package (desalted crude)	vendor package	185
X-103	Neutraliser / filming amine package (atm OH)	vendor package	185
X-104	Corrosion inhibitor package (VDU OH)	vendor package	185

5 Accuracy and next steps

The heaters, the column shells and internals, and the hot exchanger bank account for more than 60 % of equipment cost. Vendor budget quotes for H-101, H-201, C-101, C-201 and the desalters should be obtained to move the estimate to Class 3.